

Computer Architecture Lab

(EL-2012)

LABORATORY REPORT

Fall 2026



(Lab#3)

Apply Behavioral-Level Verilog to Design and Simulate Combinational Modules

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Date performed: 5th September, 2026

MARKS AWARDED: _____ / 10

Signature and date of Instructor: _____

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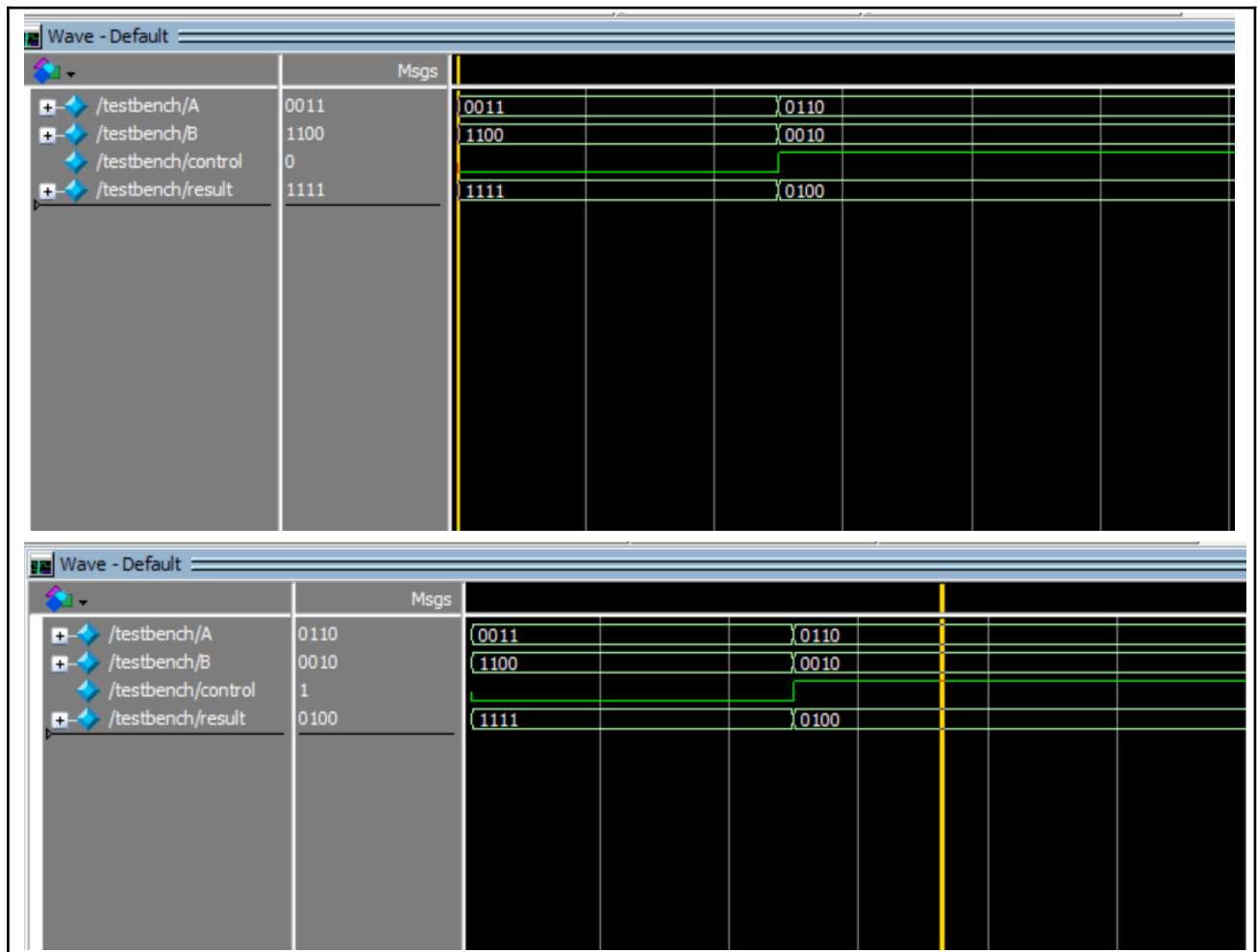
NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES, ISLAMABAD

Prepared by: Fasih Ahmad

Version: 1.0

Task 1:

Description:
Wrote Verilog code for a 4 bit adder/subtractor which performs addition or subtraction depending on the control signal, the circuit was then simulated using testbench
Code:
<pre>module addsub(inputA,inputB,ctrl,out); input [3:0] inputA; input [3:0] inputB; input ctrl; output reg [3:0] out; always @* begin if (ctrl==1) out=inputA-inputB; else out=inputA+inputB; end endmodule module testbench(); reg [3:0] A; reg [3:0] B; reg control; wire [3:0] result; addsub uut(A,B,control,result); initial begin A=4'b0011; B=4'b1100; control=0; #50; A = 4'b0110; B = 4'b0010; control = 1; #50; end endmodule</pre>
Output:



Task 2:

Description:

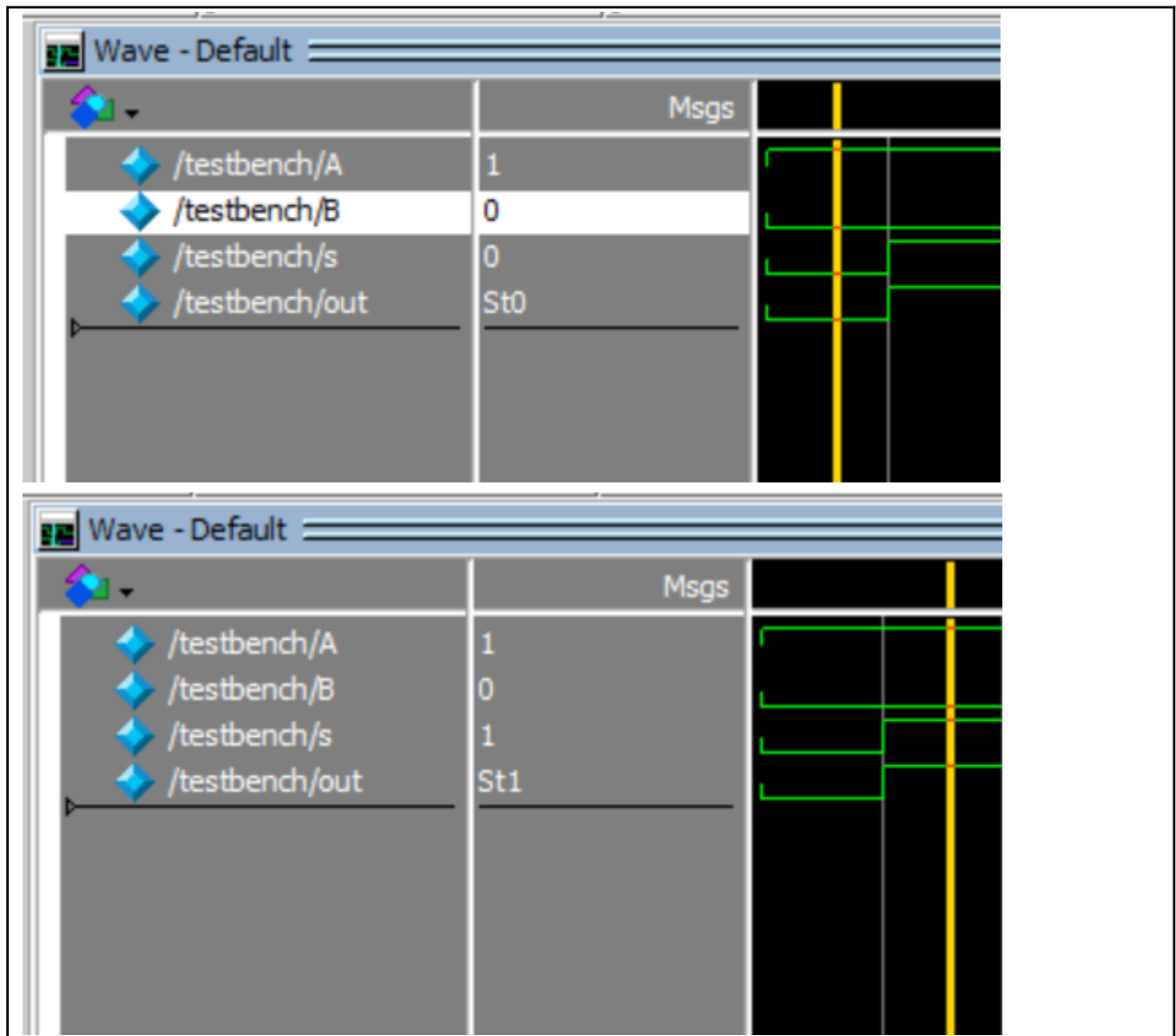
We wrote verilog code for a 2x1 MUX having 1 bit inputs and outputs , which performed logical AND when select line was set as 0 and Logical OR when select line was 1

Code:

```
module mux(inputA,inputB,sel,out);
input inputA;
input inputB;
input sel;
output reg out;
always @*
begin
if(sel==0)
out=inputA & inputB;
else
out=inputB | inputA;
end
endmodule

module testbench();
reg A;
reg B;
reg s;
wire out;
mux m(A,B,s,out);
initial
begin
A=1;
B=0;
s=0;
#50;
A=1;
B=0;
s=1;
#50;
end
endmodule
```

Output:



Task 3:

Description:

Here we wrote a verilog code for 2 bit comparator that compares two 2-bit binary numbers and outputs result as greater than or less than or equal, the circuit was also simulated using test bench.

Code:

```
module comparator(A, B, greater, less, equal);
input [1:0] A;
input [1:0] B;
```

```
output reg greater;
output reg less;
output reg equal;
always @*
begin
    if (A > B)
    begin
        greater = 1;
        less = 0;
        equal = 0;
    end

    else if (A < B)
    begin
        greater = 0;
        less = 1;
        equal = 0;
    end

    else
    begin
        greater = 0;
        less = 0;
        equal = 1;
    end
end
endmodule
module testbench();
reg [1:0] A;
reg [1:0] B;
wire greater;
wire less;
wire equal;
comparator c(A, B, greater, less, equal);
initial
begin
A = 2'b11;
B = 2'b01;
#50;
A = 2'b01;
B = 2'b10;
#50;
A = 2'b10;
B = 2'b10;
#50;
end
```

```
endmodule  
...
```

Output:

